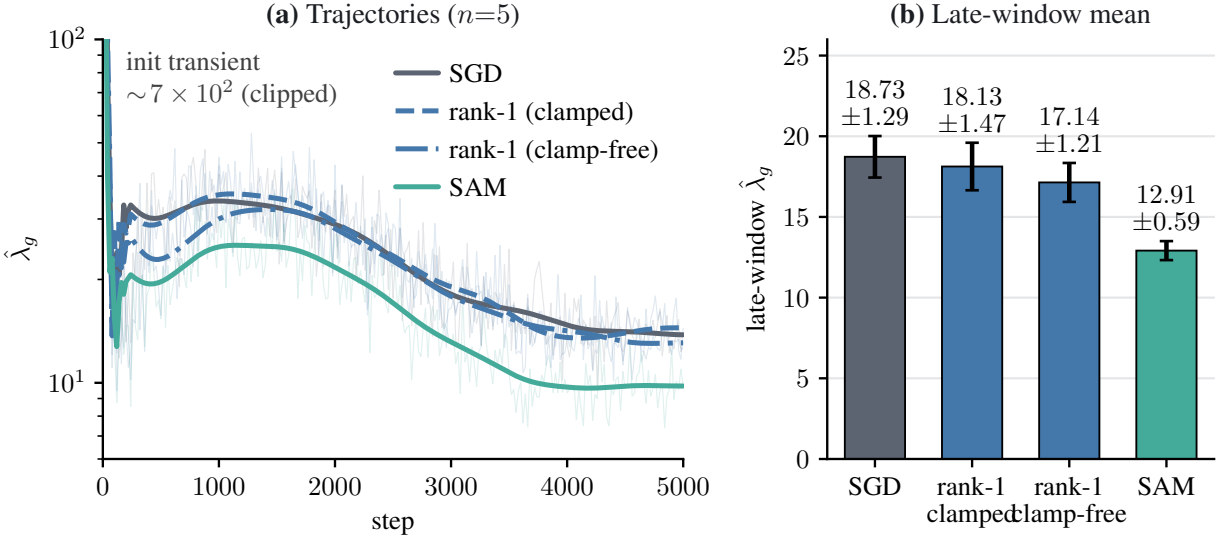

(a) Trajectories (n=5)
10^2
10^1
$\hat{\lambda}_g$
init transient
$\sim 7 \times 10^2$ (clipped)
SGD
rank-1 (clamped)
rank-1 (clamp-free)
SAM
0
1000
2000
3000
4000
5000
step
(b) Late-window mean
late-window $\hat{\lambda}_g$
25
20
15
10
5
0
18.73
±1.29
18.13
±1.47
17.14
±1.21
12.91
±0.59
SGD
rank-1 clamped
rank-1 clamp-free
SAM